

RAPHAEL ANAADUMBA

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TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C++, Java, MATLAB, R

Machine Learning: PyTorch, PyTorch Geometric, TensorFlow, Scikit-Learn, XGBoost, Keras

Graph & Geospatial ML: GNNs, GAT, GCN, GraphSAGE, NetworkX, GeoPy, ArcGIS/QGIS

Data & Engineering: Pandas, NumPy, Docker, Git, REST APIs, Node.js

Research Areas: Spatial ML, Environmental Health AI, Medical Audio ML, Privacy-Preserving ML, Signal Processing

RESEARCH & APPLIED ML EXPERIENCE

Graduate Research Assistant | CUBICS Lab, UMass Lowell

Jan 2023 – Present

Lowell, MA

- Lead developer of **NSF-funded (\$2.5M)** GNN frameworks for municipal lead contamination risk prediction, validated across **50,000+ water test records** spanning Chicago, Flint MI, and Massachusetts
- Designed spatial graph models integrating Census, ACS, and EPA datasets to support transferable risk assessment in cities with limited local testing data
- Built privacy-aware ML workflows for partially masked public-sector address records supporting lead-risk assessment under evolving **EPA Lead and Copper Rule Improvements (LCRI, 2027 enforcement)**
- Built reproducible ML workflows covering data ingestion, preprocessing, feature normalization, graph construction, model training, ablation testing, threshold optimization, and GIS-ready output generation.
- Published **6 peer-reviewed papers** across AAAI, IEEE/ACM CHASE, ICASSP, Frontiers in Environmental Engineering, and Nature Scientific Reports presented at AAAI 2026 and IEEE/ACM CHASE 2025

SELECTED PUBLICATIONS

Spatial GAT for Neighborhood-Scale Lead Contamination Risk Prediction — First Author | AAAI 2026 2024–2025

Python, PyTorch Geometric, GAT, XGBoost, GeoPy, ArcGIS

- Aggregated 38,000+ EPA water tests to 739 census block groups and built a directed hybrid spatial graph using GAT embeddings, achieving 91.6% unsafe recall and AUROC of 0.83, with all 10 predicted hotspots located adjacent to confirmed unsafe zone.
- Engineered learnable edge-weight formula combining vulnerability gradient, spatial proximity, and demographic similarity; produced risk maps exportable directly to ArcGIS/QGIS for municipal operations

Dementia Detection from Speech: AST + GAT — First Author | ICASSP 2026

2024–2025

Python, PyTorch, Audio Spectrogram Transformer, GAT, Pitt Corpus, TAUADIAL

- Fused acoustic spectrograms with relational speaker-utterance graphs via AST+GAT; achieved **80.5% accuracy on Pitt Corpus (9.7% above AST-only)** and **81.8% on TAUADIAL (30.5% improvement)**; ablation confirmed temporal GAT edges as primary MCI detection driver
- Designed for telehealth deployment — non-invasive, low-resource, requires only conversational audio; supporting scalable research toward low-resource dementia screening

Respiratory Disease Classification via GCN — First Author | IEEE/ACM CHASE 2025

Python, PyTorch Geometric, GCN, Signal Processing, Audio Synthesis

- Built lung-sound analysis system using GCN over synthesized non-speech audio features, validated on **1,000+ clinical recordings** across multiple respiratory conditions; designed for lightweight deployment on edge devices in rural clinics
- Introduced graph-based relational modeling to respiratory audio analysis, treating clinical recordings as structured graphs rather than flat feature vectors, improving classification robustness across disease subtypes

GNN-Based Water Contamination Detection — Flint, MI — First Author | Frontiers in Environmental Engineering, 2025

Python, GAT, GCN, Census API, Selenium, GeoPy

- Identified household-level lead risk across **14,000+ Flint homes** using only public housing and demographic data — achieved **0.93 AUC** without household water sampling, demonstrating real-world deployability
- Developed privacy-compliant data pipeline (Selenium) for Massachusetts EEA lead portal and Census API integration, enabling cross-city transferability without exposing personally identifiable address records

Self-Supervised GAT for Community-Engaged Lead Contamination Risk Assessment

2026

Nature Scientific Reports — First Author | Accepted, In Press

- Introduced self-supervised learning to spatial GNN-based environmental risk modeling, reducing dependence on labeled water testing data and supporting deployment in municipalities where ground-truth sampling is limited
- Extending community-engaged lead risk assessment to cities with no prior testing infrastructure; accepted as first author in **Nature Scientific Reports** (2026, In Press)

SOFTWARE ENGINEERING EXPERIENCE

Software Engineer | Shenzhen Roynard Service Consulting Co.

May 2021 – Nov 2022

Shenzhen, China

- Developed Node.js backend services and REST API integrations for client-facing web platforms, supporting authentication, data handling, and business workflow automation across multiple consulting projects
- Debugged production issues, collaborated with cross-functional teams, and maintained backend features with version control and modular API design to improve maintainability and delivery reliability
- Applied database-connected service logic and modular architecture patterns to support scalable, maintainable backend systems across diverse client environments

Software Engineer Intern | National Health Insurance Authority

May 2015 – Aug 2015

Accra, Ghana

- Maintained and optimized health insurance card-printing systems serving vulnerable populations, improving processing reliability across national operations

Computer Science Teacher & Lab Technician | Tamale Senior High School

Sep 2016 – Aug 2018

Ghana

- Taught programming fundamentals (C, C++) and administered wired/wireless network infrastructure supporting daily operations for **700+ students**

EDUCATION

PhD, Computer Science | University of Massachusetts Lowell

Jan 2023 – Present

Research: Graph Neural Networks for Environmental Health & Medical Diagnostics | **NSF Award #2230180**

MSc, Computer Science | Nanjing University of Information Science & Technology

Sep 2018 – Jun 2020

Specialization: Machine Learning | Nanjing, China

BSc, Computer Science | University for Development Studies

Sep 2012 – Jun 2016

Tamale, Ghana

SERVICE & RECOGNITION

Invited Peer Reviewer | Springer Nature

2024 – Present

- Certified reviewer for **3 Springer Nature journals**: Scientific Reports, The Journal of Supercomputing, and Discover Sustainability — **4 completed reviews** (2024–2026), 5th in progress; awarded official Springer Nature Reviewer Certificates

NSF Student Travel Award — Competitively selected to present at IEEE/ACM CHASE 2025, New York

2025

NSF-Funded Research Contributor — Core researcher on \$2.5M NSF-funded environmental ML project (Award #2230180) 2023